In [6]:
```
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Load Iris dataset
iris = load_iris()

# Create DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)

# Independent variable (X)
X = df[['sepal length (cm)']]

# Dependent variable (y)
y = df['petal length (cm)']

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Print coefficient and intercept
print("Coefficient (Slope):", model.coef_[0])
print("Intercept:", model.intercept_)

# Evaluate model
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R² Score:", r2_score(y_test, y_pred))

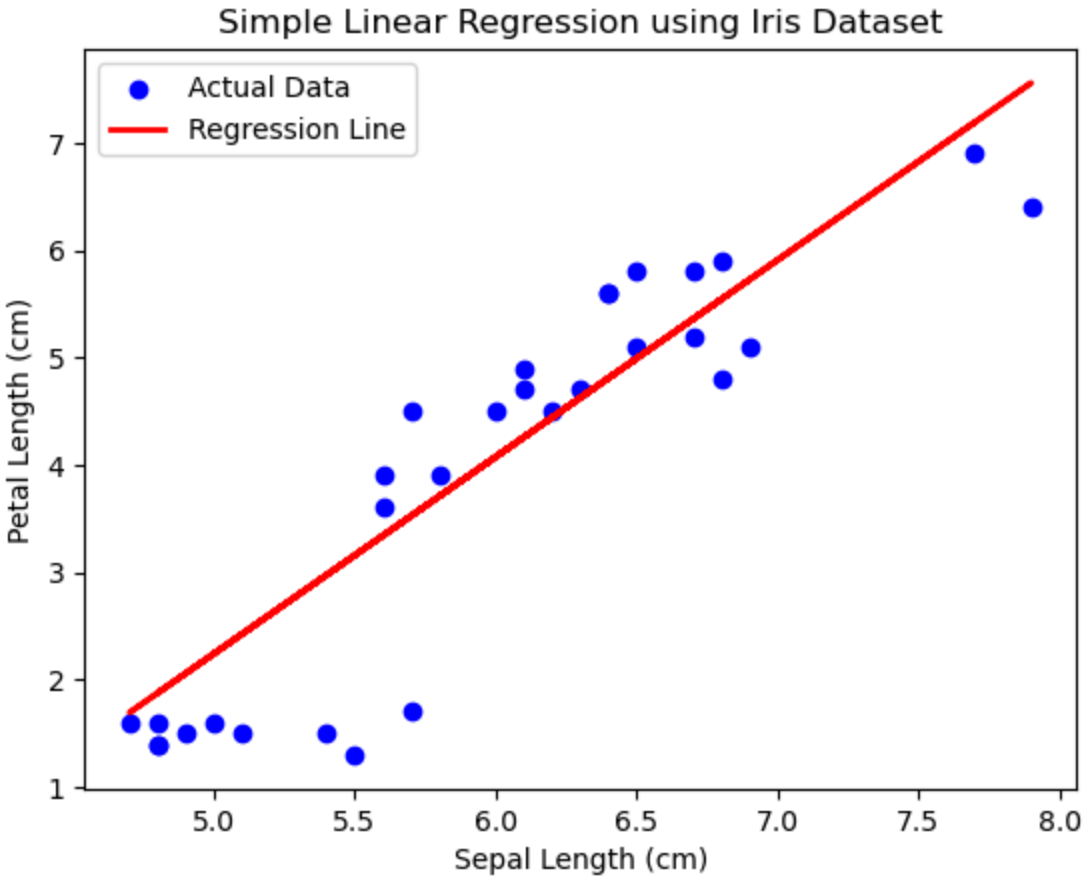
# Plot actual data points
plt.scatter(X_test, y_test, color='blue', label='Actual Data')

# Plot regression line
plt.plot(X_test, y_pred, color='red', linewidth=2, label='Regression Line')

# Labels and title
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Petal Length (cm)')
plt.title('Simple Linear Regression using Iris Dataset')
plt.legend()
```

```
# Show graph
plt.show()
```

```
Coefficient (Slope): 1.8339625990203465
Intercept: -6.927127731475695
Mean Squared Error: 0.5960765879745186
R² Score: 0.8181245472591437
```



In [ ]: